

ERD Data Model Evaluation Report

Target System: Asset Management System Status: Draft for Revision

Executive Summary: The evaluated Entity-Relationship Diagram (ERD) serves as a solid foundational architecture, covering essential core entities and business flows. However, significant structural gaps remain in audit capabilities, security models, compliance tracking, and business logic enforcement that must be addressed prior to implementation.

1. WELL-DESIGNED BASELINE COMPONENTS

✓ **Core Asset Management**

Assets, Categories, Locations, Users, and Suppliers entities are effectively established.

✓ **Procurement & Lifecycle**

Includes Purchase Orders, Invoices, and Warranties for acquisition tracking.

✓ **Operational Workflows**

Covers Custody Transfers, Maintenance Templates/Work Orders, and Stocktakes.

✓ **Hierarchical Structure**

Location hierarchy is properly modeled using a recursive parent_id self-reference.

2. CRITICAL ARCHITECTURE GAPS & FLAWS

1. Missing Immutable Audit Trail (AssetStatusHistory)

COMPLIANCE RISK

Lack of an immutable status history table allows untraceable changes. Project requirements strictly demand an unalterable log of asset status transitions and historical ownership.

2. Absent Retirement & Disposal Management Entity

GOVERNANCE

There is no dedicated Retirement or Disposal entity to capture disposal reasons, sign-off approvals, decommissioning dates, or supporting documentation/proof.

3. Uncoupled Custody Tracking Architecture

DATA INTEGRITY

Relying on direct foreign keys inside the Assets table rather than an independent CustodyAssignment entity compromises historical accuracy and prevents point-in-time custody auditing.

4. Lack of Scoped Role-Based Access Control (RBAC)

SECURITY

The standard Users table lacks multi-tenant or organizational scoping mechanisms. This fails to prevent users from one college/department from accessing assets of another.

2. CRITICAL ARCHITECTURE GAPS & FLAWS (CONTINUED)

5. Insecure Document & Attachment Handling

SECURITY

Invoices and warranties rely on plain text URL strings. There is no central, secure Attachments/Documents table to govern access control and metadata for sensitive files.

6. Incomplete Maintenance Scheduling & Trigger Logic

OPERATIONS

Maintenance templates do not account for triggering by asset model, operating runtime, or condition. Additionally, there is no explicit calculation or storage for Next Due Date.

7. Flawed Stocktake Session Architecture

DATA INTEGRITY

The stocktake module does not freeze or record an expected asset baseline snapshot at session initiation, making it impossible to reliably detect or report "Missing Assets".

8. Missing Structural Constraints & Referential Integrity Rules

VALIDATION

Cardinalities and explicit constraints are missing—specifically those preventing cyclic location relationships (a room being its own parent) and ensuring invoices tie directly to valid purchase orders.

3. CONCLUSION & RECOMMENDATIONS

Final Verdict: Good Foundation, Requires Redesign

The current ERD is a promising starting point for basic asset tracking. However, fundamental design enhancements must be implemented before it can be considered compliant with enterprise project requirements. Priority focus areas for the next revision must include **Audit History**, **Asset Retirement Workflow**, **Scoped RBAC**, **Secure Document Storage**, and **Dynamic Maintenance Scheduling**.